Source-Assisted Hierarchical Semantic Calibration Method for Ship Detection Across Different Satellite SAR Images

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***Abstract*— With the increase of spaceborne synthetic aperture radar (SAR) platforms, numerous SAR images are available for ship detection applications. Traditional deep learning-based detection methods struggle with the distributional disparities in SAR images acquired from different platforms, arising from differences in radar characteristics and data acquisition conditions. Existing approaches employ domain adaptation (DA) techniques to align domain distribution and thus mitigate distribution divergence. However, due to the inherent specificity of SAR images, i.e., ship targets and background environments exhibit highly visual similarity, these methods may inadvertently destroy the discriminative representations of ship targets, resulting in poor cross-domain detection performance. To alleviate this dilemma, we propose a source-assisted hierarchical semantic calibration (SHSC) framework for ship detection across different satellite SAR images. First, a source-assisted semantic calibration module (SSCM) is designed, which performs multilevel semantic calibration by constructing a source-assisted (SA) detector as a guiding mechanism to preserve the discriminative semantics of ship targets. Then, the uncertainty-aware guided feature-level alignment module (UG-FAM) and instance-level alignment module (UG-IAM) are developed, which effectively capture the crucial ship target attributes by emphasizing the learning of those discriminative samples. Extensive experiments are conducted on the datasets obtained from the TerraSAR, Gaofen-3, Sentinel-1, and RadarSat-2 satellites. The experimental results show that the proposed SHSC method outperforms the other UDA approach by an average of more than 3% on AP in ship target detection accuracy across different satellite SAR images.**

***Index Terms*— Crossing different satellites, domain adaptation (DA), semantic calibration, synthetic aperture radar (SAR) ship detection.**

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I. Introduction

SYNTHETIC aperture radar (SAR) is a remote sensing technique that can achieve high-resolution imaging of terrain by emitting a series of microwave pulse signals and receiving the reflected signals [1]. It can be deployed on various platforms, including aircraft, satellites, and spaceships. SAR has become an important tool for ocean monitoring and resource management because it is not limited by weather, light conditions, or occlusions and is capable of all-day and all-weather observations [2]. SAR ship detection [3], as an essential application of SAR technology, aims to achieve real-time monitoring and rapid response to maritime activities by accurately identifying and locating ships in SAR images, thus providing crucial information for marine management.

Numerous algorithms have been developed, given the practical significance of SAR ship detection tasks. These methods can be roughly categorized into traditional and deep learning-based approaches. Traditional detection approaches primarily rely on the physical characteristics of the ship targets, such as scattering characteristics [4], [5], [6], [7], [8], [9], polarization characteristics [10], [11], [12], [13], [14], [15], and geometric characteristics [16], [17], [18], to distinguish the ship and background. This approach usually performs well in scenarios with obvious physical characteristics, relatively low complexity, and noise. However, it has limitations when dealing with complex, high-noise, and diverse target scenarios. The deep learning-based detection methods have demonstrated remarkable advantages in SAR ship detection [19], [20], due to their powerful feature extraction ability and adaptability to complex scenes and multiscale targets. It aims to automatically learn the feature representations of ship targets in SAR images by utilizing a deep convolutional neural network (DCNN) model [21], and learned features are further subjected to classification and bounding box regression to achieve the ship detection task.

With the increasing deployment of spaceborne SAR platforms, more and more SAR images can be acquired for SAR

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ship detection. Nevertheless, due to the varying technical specifications and system parameters among different SAR platforms, such as radar frequency, beamwidth, orbit altitude, observation angle, and so on, the imaging data of different SAR platforms generally present different probability distributions. This heterogeneity poses a challenge for traditional deep learning-based SAR ship detection algorithms, which typically depend on the assumption that the source domain (i.e., training data) and the target domain (i.e., test data) are identically and independently distributed (i.i.d.). When a model trained on one SAR platform image is applied to another, the detection performance of the model will be significantly degraded. It is also called the *domain shift* problem. The impact of domain difference can be observed in Fig. 1, where a ResNet101 [22] based faster R-CNN [23] model trained on SAR images from Gaofen-3 satellite, and tested on SAR images from the Sentinel-1 satellite. The results presented in the second row in Fig. 1 validate our idea that a significant performance drops with false detection when the domain discrepancy is present. A straightforward solution is to develop specific detection models for different spaceborne SAR platforms. However, this approach is impractical due to the inherent difficulties associated with annotating SAR images. The lack of visual details, significant background interference, and unclear ship contours make the annotation process time-consuming and labor-intensive. Therefore, developing a more robust and adaptive solution that can handle domain shifts effectively is crucial for SAR ship detection.

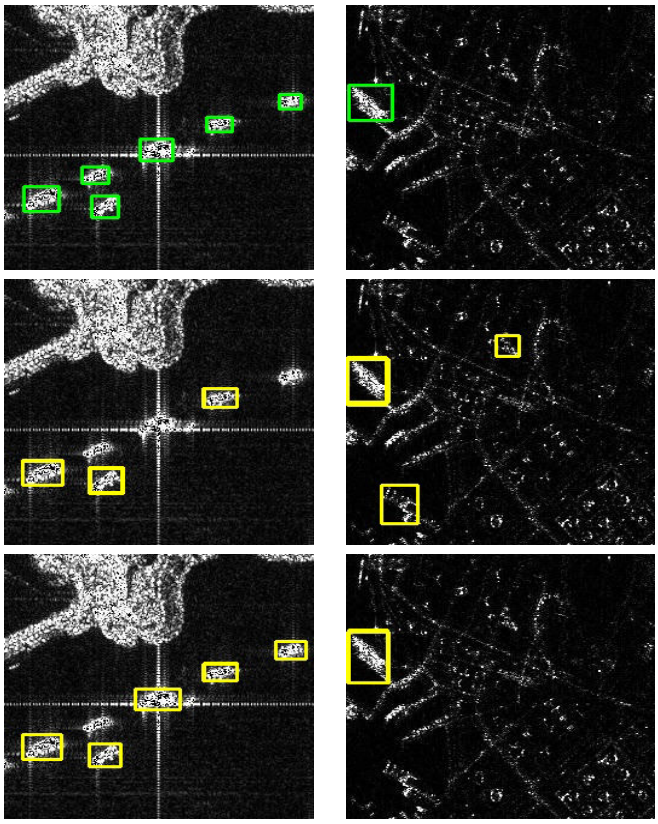


Fig. 1. Example of ship detection in cross-domain SAR images. The left column is the SAR images from the Gaofen-3 satellite, and the right is the SAR images from the Sentinel-1 satellite. The first row denotes the ground truth. The second row shows the results predicted using the traditional faster R-CNN trained on SAR images from the Gaofen-3, and we can see that there is a serious misdetection on the domain-shifted Sentinel-1 satellite SAR images. The third row presents the results of our method, which can effectively eliminate the domain discrepancy between them.

Unsupervised domain adaptation (UDA) [24] techniques provide a solution to improve the model performance on the target domains by migrating knowledge learned from the labeled source domains to the completely unlabeled target domains with different distributions. In cross-domain SAR ship detection, most UDA-based methods primarily employ a domain adversarial training (DAT) [25] strategy. DAT aims to minimize the domain gap by aligning the distributions of the source and target domain, thereby achieving effective domain adaptation (DA) [26]. The core principle of DAT involves an adversarial training process between a generator and a domain discriminator. The discriminator is trained to distinguish the domains, while the generator is simultaneously trained to fool the discriminator, making it difficult to distinguish which domain the features come from. Through such adversarial, the model learns to extract domain-invariant representations [27] (i.e., features are shared across all domains). These representations bridge the domain differences, thus facilitating cross-domain prediction. For instance, in [28], an adversarial learning attention module is used to align features at both the pixel-level and the instance-level, and the compensation loss module is utilized to perform texture compensation for candidate object region at the instance level, further reducing the domain discrepancies. Zhao et al. [29] proposed a feature decomposition-based approach to decompose the features of the source and the target domain into domain-invariant and domain-specific representations. These domain-invariant features are then fused to enhance global feature extraction. Furthermore, Shi et al. [30] introduced a comprehensive approach that combines pixel and multilevel feature domain adaptation to align features at the pixel-level, feature-level, and instance-level. They also incorporate an iterative pseudo-labeling [31] strategy that generates and leverages pseudo-labels progressively to facilitate the detector learning on the target domain. Shi et al. [32] proposed a progressive transfer SAR ship detection approach, which gradually adapts the optical domain to the SAR domain at the pixel-level, feature-level, and prediction-level. Similarly, Xu et al. [33] introduced a multilevel alignment model, which integrates image-level, convolution-level, and instance-level alignment to promote detector adaptation from a labeled optical image to an unlabeled SAR image.

Nevertheless, despite these methods largely reducing distribution discrepancy between different domains through multilevel adversarial learning, they ignore the unique characteristics of SAR images. SAR images, distinct from natural images, usually exhibit similar scattering intensity between ship targets and background surroundings. This characteristic is manifested in the ship targets and the background surroundings having similar visual appearance and texture. During the adversarial domain adaptation, the feature distribution of the source domain gradually closes to the distribution of the target domain, thus achieving domain distribution alignment. However, since target domain images are unlabeled, their feature

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distribution in the feature space becomes highly chaotic. This indiscriminate domain alignment may inadvertently disrupt the discriminative representation of ship targets learned from the source domain, ultimately leading to a degradation in cross-domain detection performance.

Considering the problem of discriminative corruption of ship targets during the adversarial domain adaptation caused by the inherent specificity of SAR images, we propose a source-assisted hierarchical semantic calibration (SHSC) framework for ship detection across SAR images from different satellites. It aims to better preserve the discriminative representation of ship targets during domain adaptation, thus enhancing the cross-domain detection performance. Specifically, we propose a source-assisted semantic calibration module (SSCM), which performs multilevel semantic calibration to preserve the discriminative attributes of ship targets. This is achieved by constructing a source-assisted (SA) detector and leveraging the knowledge distillation (KD) mechanism to effectively transfer the feature representation from the source domain into the domain adaptation process. In the following, based on adversarial training, the uncertainty-aware guided feature-level alignment module (UG-FAM) and instance-level alignment module (UG-IAM) are proposed. These modules are designed to strengthen the learning of those discriminative samples by reweighting the domain representations, thus fully capturing the critical semantics associated with ship targets. With multiple hierarchical semantic calibrations, UG-FAM and UG-IAM, the model can better preserve the discriminative semantics of ship targets while adapting to diverse domains. We perform extensive experiments on the SAR ship detection datasets acquired by Sentinel-1, TerraSAR, Gaofen-3, and RadarSat-2 satellites. The experiment results consistently demonstrate the remarkable advantages of the proposed SHSC framework in SAR cross-domain ship detection performance compared to other methods. In addition, detailed model analysis experiments further validate the effectiveness of the SSCM, UG-FAM, and UG-IAM components in improving model discriminability and adaptability.

In summary, our contributions are as follows.

1) Aiming at the issue of the discriminative corruption of ship targets during the adversarial domain adaptation process induced by the inherent specificity of SAR images, a SHSC method is proposed, to better preserve the discriminative representations of ship targets, thus improving the cross-domain detection performance.
2) A multiple hierarchical semantic calibration mechanism is proposed to gradually preserve the discriminative semantics of ship targets during domain adaptation. This is reflected in 1) SSCM, which is designed to perform multilevel semantic calibration by introducing additional source representations as guidance. 2) UG-FAM and UG-IAM are developed to enhance the learning of discriminative samples. Through such multilevel semantic calibration, the model can adapt to different domains while preserving the discriminative of ship targets.
3) Extensive experiments are conducted on the SAR ship detection datasets acquired by Sentinel-1, Gaofen-3, TerraSAR, and RadarSat-2 satellites. The experimental results show that the proposed SHSC framework surpasses other UDA approaches by over 2% in ship target detection accuracy across diverse satellite SAR images. Furthermore, in-depth empirical analysis experiments confirmed the effectiveness of the proposed SSCM, UG-FAM, and UG-IAM components in enhancing cross-domain SAR ship detection performance.

The remaining parts of the article are organized as below. The related work is discussed in Section II. The proposed SHSC is described in Section III. The experimental results and model analysis are presented in Section IV. Finally, Section V concludes this article.

II. Related Work

A. Deep Learning-Based SAR Images Object Detection (OD)

The deep learning-based OD method [34], [35], [36], [37] aims to automatically identify and localize targets in images or videos through DCNN. With powerful feature extraction capabilities and adaptability to complex scenes and multiscale targets, deep learning has made remarkable progress in SAR applications. SAR ship detection methods based on deep learning can be broadly categorized into one-stage and two-stage detection approaches. Among them, the one-stage detection algorithm aims to directly predict the location and class of the object from image features without generating candidate object regions. Classic methods include YOLO series [38], SSD [39], and RetinaNet [40]. For example, Yang et al. [41] proposed a super lightweight and efficient SAR ship detection method based on the YOLOv5 architecture. Han et al. [42] enhanced the representation of small ships in the feature pyramid by utilizing deconvolution, thus improving the detection accuracy of SSD. The two-stage detection approach involves two main steps: 1) generating potential candidate object regions from the input images through the region proposal network (RPN), and 2) using the classifier to classify the ship targets for each candidate region, and the regressor to fine-tune the position of the bounding box. The representative approaches include faster R-CNN [23], mask RCNN [43], and so on [44]. Due to the scalability and performance advantages of faster R-CNN, most of the SAR ship detection methods are implemented based on it [45]. For instance, based on faster R-CNN, Lin et al. [46] introduced squeeze and excitation mechanisms to improve ship detection performance. Wei et al. [47] proposed a parallel high-resolution FPN [48] that fully integrates and utilizes multiscale feature maps for SAR ship detection. While these algorithms demonstrated superior detection performance in various scenarios, their performance typically depends on the assumption that training and testing data have the same distribution. This assumption is often violated in real-world scenarios due to the discrepancy in technical parameters and data acquisition conditional between different SAR imagery platforms, such as operating frequency, resolution, polarization mode, and geometric observation conditions. It is also known as the domain shift problem and can severely deteriorate model performance. In this article, we focus on the cross-domain SAR ship detection task, aiming to reduce the distribution dif-

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ference between images from different SAR imaging platforms to achieve more effective cross-domain ship detection.

B. UDA for OD

In practical applications, traditional deep learning-based OD algorithms are often limited by training data and struggle to adapt to changes in the external environment. To improve the adaptability of the detection models, the UDA-based OD (UDAOD) method has gained attention. UDA technique aims to migrate knowledge to unlabeled target domains with different distributions, leveraging fully labeled source domains to improve model performance on the target domains. Its core idea is to minimize domain distribution differences and learn domain-invariant representations. As the earliest cross-domain OD algorithm, domain adaptive faster R-CNN (DAF) [49] employs an adversarial training strategy based on the gradient reversal layer (GRL) [24], which aligns domain distribution at the image level and instance level, thus reducing domain differences. Subsequently, multiadversarial faster R-CNN (MAF) [50] adapts a multilevel adversarial training strategy to perform domain alignment operations in different convolutional layers. Khindkar et al. [51] introduced a ResIduaL seLf-attentive featUre alignMEnt (ILLUME) method, incorporating a self-attention feature map module to enhance the structural attention to the object-related region, thereby generating domain-invariant features. Chen et al. [52] proposed a hierarchical transferability calibration network (HTCN) approach to balance the transferability and discriminability of the model. Shen et al. [53] proposed a stacked complementary losses (SCLs) approach, which applies gradient detach operations to complement information on feature representations at different network stages. Considering that different hierarchical features have different transferability, Saito et al. [54] proposed a strong-weak domain alignment (SWDA) model that integrates weak global alignment and strong local alignment to improve the cross-domain adaptability of the model.

In the context of cross-domain SAR ship detection, Shi et al. [32] proposed a progressive transfer UDA framework for SAR ship detection by migrating knowledge from the optical images to the SAR images. In addition, Zhao et al. [28] proposed a cross-different satellite SAR ship detection approach by using feature alignment and compensation loss. Zhao et al. [29] proposed a feature decomposition-based method by decomposing the features of source and target domains into domain-invariant and domain-specific features. Zhao et al. [55] introduced the domain-specific knowledge for cross-domain SAR image ship detection. Xu et al. [33] employed a multilevel alignment strategy to align feature distribution at the image levels, convolutional levels, and instance levels to adapt the detector from labeled optical domains to unlabeled SAR domains. These approaches significantly reduce the distribution difference among different data and provide effective technical means for cross-domain SAR ship detection. However, due to the particularity of SAR images, it is necessary to consider preserving the discriminative representation of the ship target itself while adapting across domains. In this article, we work on this issue and propose an SA hierarchical semantic calibration method to better preserve the discriminative features of ship targets while adapting to different domains.

III. Proposed Method

This section provides the technical details of the proposed SHSC approach. The overall framework of the proposed SHSC is illustrated in Fig. 2. The primary objective of SHSC is to effectively preserve the discriminability of ship targets while adapting to different satellite SAR images, thus improving cross-domain ship detection performance. Based on this, the proposed SHSC comprises three key modules: the SSCM detailed in Section III-C, the UG-FAM and UG-IAM introduced in Section III-D. The problem formulation is introduced in Section III-A, the network overview is presented in Section III-B, and the whole model optimization objective is detailed in Section III-E.

A. Problem Formulation

For cross-domain SAR ship detection tasks, it is needed to predict both the target class and the boundary box location. Formally, we denote a fully labeled source data as $\mathcal{D}_s = \{(x_i^s, y_i^s, b_i^s)\}_{i=1}^{N_s}$ comprising N_s samples. Here, x_i^s represents source training samples, $b_i^s \in \mathcal{R}^{\kappa\times 4}$ represents a list of bounding boxes, where κ is the number of boxes, and each of which is expressed by 4-D coordinates. $y_i^s \in \mathcal{R}^{\kappa\times 1}$ denotes corresponding class labels for one box. Similarly, we denote the unlabelled target data $\mathcal{D}_t = \{x_j^t\}_{j=1}^{N_t}$, where N_t is the number of target domain data. The source and target domains have the same label space but violate the i.i.d. assumption because they are acquired from different satellite SAR platforms. This article aims to learn an adaptive ship target detector using labeled D_s and unlabelled D_t, such that the detector performs well on the target domain.

B. Network Framework Overview

As mentioned in Section I, detection models are susceptible to differences in data distribution. On the other hand, SAR images present unique challenges in cross-domain target detection due to the similar visual appearance and texture between ship targets and the background interference. When employing the DAT strategy, the distribution of the source domain gradually approaches the distribution of the target domain, thus achieving domain alignment. However, this indiscriminate domain alignment may inadvertently destroy discriminative representations of ship targets learned from the source domains. This further leads to the learned adaptive detector failing to accurately predict the target domain images.

To address these challenges, we propose a SHSC method for cross-domain SAR ship target detection. The faster R-CNN [23] is used as the basic detector. Following the mainstream cross-domain detection method [49], the feature-level and instance-level domain alignments are also performed. However, considering the characteristics of SAR images, with substantial improvements for enhancing the discriminative learning of the model. In particular, an SSCM is designed for

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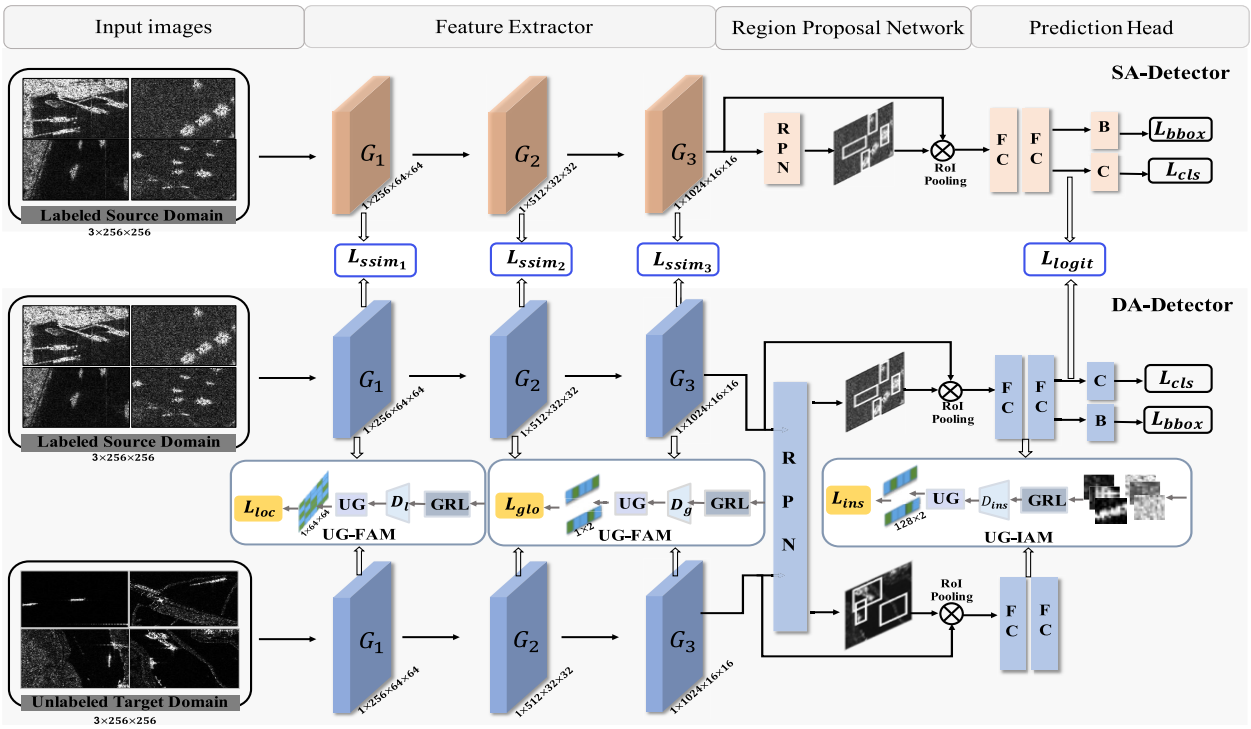


Fig. 2. Overall architecture of the proposed SHSC. SHSC comprises a SA detector and a DA detector, both based on faster R-CNN, where G_1, G_2, and G_3 are three convolutional blocks. The SA detector performs multilevel semantic calibration for preserving discriminative features with L_{ssim} at feature levels and L_{logit} at the decision level. The DA detector focuses on distributional alignment between source and target domains at feature and instance levels. The uncertainty-aware guided (UG) strategies are designed and used in UG-FAM and UG-IAM. These facilitate the model to adequately capture the ship object's discriminative attributes during adversarial adaptation.

multilevel semantic calibration to preserve the discriminative features. Furthermore, the UG-FAM and UG-IAM are developed to promote the model to fully capture the discriminative semantics related to the ship targets. In the following, we will briefly review the faster R-CNN method, introduce our two new modules, and summarize the model.

Faster R-CNN: The faster R-CNN model, shown in the gray area of Fig. 2, consists of a feature extractor, a RPN and a prediction head. The feature extractor, defined as G, typically contains several convolutional blocks and activation layers (e.g., ResNet [22]). In our method, the feature extractor G is decomposed into three blocks, i.e., $G = G_1, G_2, G_3$, to facilitate the introduction of our added components. Here, G represents all three blocks. The image x is fed into the feature extractor G and yields the feature map $G(x)$. Subsequently, $G(x)$ is fed as input into the RPN network R to produce a set of region proposals. These proposals are then used to extract instance-level features from the feature maps using ROI-pooling. For brevity, all instance-level features obtained after ROI pooling are denoted as $R(G(x))$. Finally, the instance-level features are sent to the prediction head to predict class labels and bounding boxes. The training loss for the prediction head can be formulated as follows:

$$L_{\text{det}} = \frac{1}{n_s}\sum_{i=1}^{n_s}\left(L_{\text{cls}}\left(R\left(G\left(x_i^s\right)\right), y_i^s\right) + L_{\text{bbox}}\left(R\left(G\left(x_i^s\right)\right), b_i^s\right)\right) \tag{1}$$

where L_{cls} and L_{bbox} represent the classification and localization loss, respectively. For simplicity, we omit the discussion of the RPN loss. However, it is included in the training process, consistent with the original faster R-CNN model. Note that the target domain is unlabeled, the loss is computed only on the source domain data during training.

C. SA Semantic Calibration Module

The SSCM is designed to preserve discriminative representations of ship targets during adversarial domain adaptation. The description of the SSCM can be found in Fig. 3. The design motivation stems from the fact that the target domain is unlabeled, while the source domain with available labels enables the model to learn critical semantic information associated with ship targets through supervised learning. Therefore, incorporating supervised source representations can help preserve discriminative information about ship targets during adversarial adaptation. To achieve this, the KD mechanism [56] is used, which aims to transfer knowledge from a teacher model to a student model by using the output of the teacher model as an additional supervisory signal. Based on this, an SA detector is built as the teacher model and a DA detector as the student model. Note that both structures are identical, but the weights are not shared. The intermediate representations $\mathcal{M} \in \mathbb{R}^{C,H,W}$ of the feature extractor G from the SA detector and $\mathcal{N} \in \mathbb{R}^{C,H,W}$ from the DA detector are selected to perform feature-based KD, which can be formulated as

$$\mathcal{L}_{\text{feat}_i} = \frac{1}{N}\sum_{h=1}^{H}\sum_{w=1}^{W}\sum_{c=1}^{C}\mathcal{L}_{\varepsilon}(\mathcal{M}_{h,w,c}, \mathcal{N}_{h,w,c}) \tag{2}$$

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where C is the number of channels, H and W are the height and width of the feature map, respectively, and N is the batch size. i denotes the semantic calibration performed on the ith feature extraction block. $\mathcal{L}_\varepsilon$ is the difference measure function. Considering the characteristics of SAR images, where the critical semantic attribute of the ship targets is mainly reflected in their structural information. To better distill the discriminative representation of the ship targets from the source domain, inspired by De Rijk et al. [57], the structural similarity index measure (SSIM) [58] is introduced as the difference measure function. SSIM is used to measure the structural similarity of two images, which reflects the structural information of the targets by considering contrast, luminance, and structure. Using SSIM as a training objective can help the model capture additional target information, which is reflected in the local mean, variance, and cross correlation relationships in the feature space of the SA and DA detectors. SSIM consists of three parts: luminance l, contrast c, and structure s, which can be defined as follows:

$$\begin{aligned} l &= \frac{2\mu_{\mathcal{M}}\mu_{\mathcal{N}} + C_1}{\mu_{\mathcal{M}}^2 + \mu_{\mathcal{N}}^2 + C_1} \\ c &= \frac{2\sigma_{\mathcal{M}}\sigma_{\mathcal{N}} + C_2}{\sigma_{\mathcal{M}}^2 + \sigma_{\mathcal{N}}^2 + C_2} \\ s &= \frac{\sigma_{\mathcal{MN}} + C_3}{\sigma_{\mathcal{M}}\sigma_{\mathcal{N}} + C_3} \end{aligned} \tag{3}$$

where $\mu_{\mathcal{M}}$ and $\mu_{\mathcal{N}}$ denote the mean, $\sigma_{\mathcal{M}}$ and $\sigma_{\mathcal{N}}$ denote the variance, $\sigma_{\mathcal{MN}}$ denotes covariance, which capture the relationship between $\mathcal{M}$ and $\mathcal{N}$. Furthermore, to ensure computational stability, $C_1 = (K_1 L)^2$, $C_2 = (K_2 L)^2$, $C_2 = (C_2/2)$, where L is the maximum value of the dynamic range of the feature map and $K_1 = 0.01$, $K_2 = 0.03$ by default. Since the range of (3) is $[-1, 1]$, it means that the higher the value, the more similar the two images are, therefore integrating the three components l, c, and s leads to the following objective:

$$\mathcal{L}_{\text{ssim}} = (1 - \text{SSIM})/2 = \left(1 - (l^\alpha \cdot c^\beta \cdot s^\gamma)\right)/2 \tag{4}$$

where α, β, and γ are default set to 1. Note that the output of the three blocks G_1, G_2, and G_3 is taken as intermediate representations for semantic calibration. The KD loss is then denoted as L_{ssim_1}, L_{ssim_2}, and L_{ssim_3} for each blocks. Therefore, the feature-based KD training objective can be written as $L_{\text{feat}} = L_{\text{ssim}_1} + L_{\text{ssim}_2} + L_{\text{ssim}_3}$.

In addition, in order to fully preserve discriminative information, the KD is performed on the prediction phase of the detectors. Specifically, given an image x_i, the logit of the ith image for the jth region proposals produced by the SA detector and the DA detector are defined as $(p_i)_j = R_{\text{sa}}(G_{\text{sa}}(x_i))$, and $(q_i)_j = R_{\text{da}}(G_{\text{da}}(x_i))$, respectively. To achieve effective KD at the logit level, the Kullback–Leibler (KL) divergence is introduced as a metric. It can measure the differences between two probability distributions. The magnitude of the KL divergence value directly reflects the significance of the gap between the distributions. The larger the KL divergence value, the more significant the gap between the distributions and vice versa. By minimizing the KL divergence between the SA detector and the DA detector at the logit level, the knowledge learned from the SA detector can be effectively transferred to the DA detector. This process can be formulated as

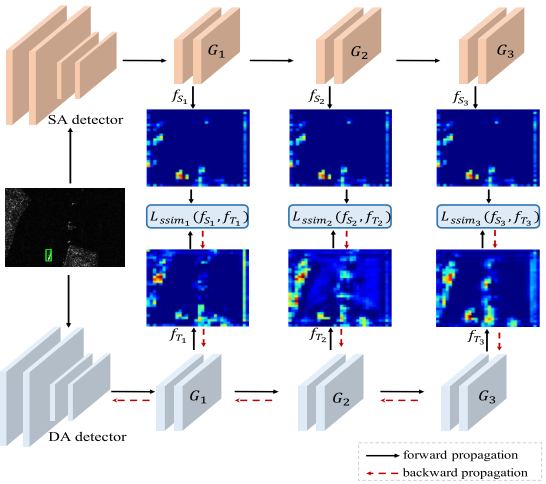


Fig. 3. Illustration of SSCM. The SA detector is built as the teacher model, and the DA detector is built as the student model. The SA detector performs multilevel semantic calibration on the feature representations f_T learned from the DA detector by using the L_{ssim}. This process facilitates the DA detector to preserve the discriminative features associated with the ship targets in the domain adaptation process.

$$L_{\text{logit}} = \frac{1}{N}\sum_{i=1}^{N_s}\sum_{j}\tau^2 \text{KL}\left(\varphi\left((p_i)_j/\tau\right), \left(\varphi\left((q_i)_j/\tau\right)\right)\right) \tag{5}$$

where $\varphi(\cdot)$ is the softmax function. τ is the temperature coefficient used to scale the smoothness of two distributions. Accordingly, the overall training objective can be written as $L_{\text{sscm}} = L_{\text{feat}} + L_{\text{logit}}$. Notably, a gradient detach operation is applied between SA and DA detectors, given their different training objectives. This operation ensures that each detector can focus on its designated task by suppressing the influence of gradient produced from semantic calibration loss during backpropagation on the SA detectors. Consequently, semantic calibration can be effectively implemented. By optimizing the minimization of L_{sscm}, the structural cues, importance features, and spatial correlation between pixels within the source representation are fully captured. These are further used to calibrate the domain adaptation representation learned from the DA-detector, thus preserving the discriminative representations of ship targets.

D. UG Alignment Module

In this section, on the basis of DAT, we propose the UG-FAM, which acts on local patch features and global image features, respectively, and the UG-IAM. The aim is to facilitate the model to adequately capture discriminative representations during adversarial adaptation. The main steps of DAT are to first map the source and target domains into a shared feature space using a feature extractor G, and then introduce a domain discriminator D based on these features to classify samples into the source or target domain. If these features have significant differences, the domain discriminator D can easily distinguish their origin. Formally, $d_i \in \{0, 1\}$ is denoted as the domain label of the ith training sample, where $d_i = 0$

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represents the source domain and $d_i = 1$ represents the target domain, then the loss for adversarial training can be written as

$$L_{\text{adv}} = \sum_x -d \log(D(G(x))) - (1-d) \log(1\text{-D}(G(x))). \tag{6}$$

In adversarial training, the discriminator D is trained to minimize the above loss, while the feature extractor G is trained to maximize it. That is, G aims to confuse the discriminator D, making it difficult to distinguish which domain the features come from. This can be realized by introducing a gradient reverse layer (GRL) between G and D, which reverses the sign of the gradients during backpropagation. Finally, adversarial training is performed by minimizing the above loss. Intuitively, if the features obtained from G cannot be easily distinguished by D, it means that similar features from the source and target domains are captured in the shared representation space. This further implies that the distribution alignment of source and target domains is achieved. However, due to the specificity of SAR images, this indiscriminate domain alignment may lead to the collapse of the discriminative representation learned from the source domain. The UG strategy is introduced to balance the model's transferability and discriminability. The core of the UG strategy is to strengthen the capture and leverage of key information by increasing the weight of distinguishable samples. By doing so, the model can better preserve the discriminative representation of the target while adapting to different domains. The UG strategy is illustrated in Fig. 4. The samples closer to the decision boundary of the domain discriminator are considered hard-to-distinguished samples, that is, they are not easily distinguished by the discriminator. Samples far from the decision boundary are considered easy-to-distinguished samples, i.e., these samples contain more discriminative information. Therefore, we strengthen the learning of easy-to-distinguished samples while relaxing the learning of hard-to-distinguished samples. This strategy is driven by two considerations: 1) enhancing learning of easy-to-distinguished samples helps the model to fully capture the discriminative features of ship targets, given their richer discriminative information. 2) Overlearning hard-to-distinguished samples may exacerbate the degradation of discriminative features, owing to the inherent properties of SAR images. The UG strategy is applied at the feature level and instance level, respectively.

1) Feature-Level Alignment: At the feature level, consider that the bottom layer features contain more local detail information, and the top layer features contain higher-level semantic information. The UG strategy is applied to local patch feature alignment at the bottom layer, i.e., G_1 and global feature alignment at the middle, i.e., G_2 and top layer, i.e., G_3. The local patch feature alignment aims to align the local regions of the feature maps, which can help maintain the consistency of local details among different domains. Global feature alignment considers the whole feature space, which is conducive to promoting consistency in high-level semantic information and reducing overall distribution differences between domains for better adaptation to the target domain.

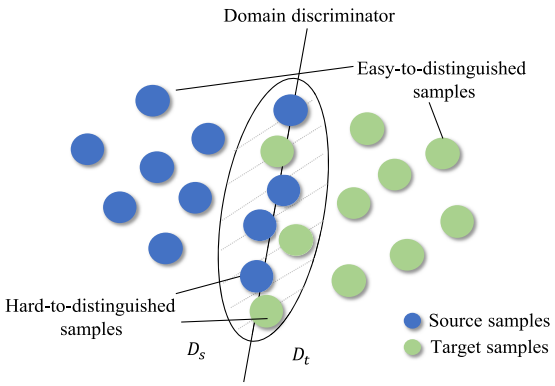


Fig. 4. Illustration of UG strategy. The blue indicates that the samples are from the source domain, and the green indicates that the samples are from the target domain. Samples close to the decision boundary of the domain discriminator in the shaded area are identified as hard-to-distinguished samples and down-weighted by our UG strategy. Samples outside the shaded area are identified as easy-to-distinguished samples that need to be up-weighted.

Specifically, first, the local discriminator D_l is designed to align the distribution on the local features. D_l is composed of several 1×1 convolutional layers. The feature map with width and height of W and H generated from G_1 are input to D_l, and corresponding domain prediction maps with the same height and width as the input features are output. Formally, given an image x_i, let $(p_i)_{w,h} = D_l(G_1(x_i))$ denote the domain prediction score at position (w, h) of the domain prediction map, where $1 \le w \le W$ and $1 \le h \le H$. Then, the uncertainty of the domain discriminator is used for an input image x_i to identify discriminative samples. The uncertainty $(g_i)_{w,h}$ of each x_i at position (w, h) from D_l can then be computed as

$$\begin{aligned}(g_i)_{w,h} = E\big((p_i)_{w,h}\big) = &-(p_i)_{w,h} \cdot \log\big((p_i)_{w,h}\big) \\ &- \big(1-(p_i)_{w,h}\big) \cdot \log\big(1-(p_i)_{w,h}\big)\end{aligned} \tag{7}$$

where $E(\cdot)$ is the entropy function. Intuitively, the less the uncertainty from D_l, the more discriminative the corresponding sample is, i.e., it is easier to be distinguished by the domain discriminator. Therefore, the samples with less uncertainty (easy-to-distinguished) by D_l should be up-weighted, and vice versa. Then, weight of each sample x_i is calculated as $2 - (g_i)_{w,h}$. Finally, the obtained uncertainty is applied to reweight the domain prediction map, which can be formulated as

$$(v_i)_{w,h} = (p_i)_{w,h} \times \big(2 - (g_i)_{w,h}\big) \tag{8}$$

where $(v_i)_{w,h}$ denotes the remodulated domain prediction map. A least-squares loss is used to train the domain discriminator D_l following [54], [59], [60]. The loss function of the local feature alignment can be formulated as

$$L_{\text{loc}_s} = \frac{1}{N_s H W} \sum_{i=1}^{N_s} \sum_{w=1}^{W} \sum_{h=1}^{H} \big(v_i^s\big)_{w,h}^2 \tag{9}$$

$$L_{\text{loc}_t} = \frac{1}{N_t H W} \sum_{i=1}^{N_t} \sum_{w=1}^{W} \sum_{h=1}^{H} \Big(1 - \big(v_i^t\big)_{w,h}\Big)^2 \tag{10}$$

$$L_{\text{loc}}(G_1, D_l) = \frac{1}{2}\big(L_{\text{loc}_s} + L_{\text{loc}_t}\big). \tag{11}$$

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Then, the global discriminator D_{g1} and D_{g2} is designed to focus on the global features at the middle and the top layer, respectively. Different from the local discriminator, the D_{g1} and D_{g2} output a single value for each input sample x_i. Let $m_i = D_{g1}(G_2(G_1(x_1)))$ and $n_i = D_{g2}(G_3(G_2(G_1(x_1))))$ be the domain prediction probability generated by D_{g1} and D_{g2}. Similar to the (7), the uncertainty from D_{g1} and D_{g2} are computed as $q_i = E(m_i)$ and $p_i = E(n_i)$. Based on computed uncertainty, the domain prediction probability is further computed as $z_i = m_i \times (2-q_i)$ and $u_i = n_i \times (2-p_i)$. The loss function of the global feature alignment at the middle and the top layer can be written as

$$L_{\text{glo}_1} = -\frac{1}{N_s}\sum_{i=1}^{N_s} d_i \log(z_i) - \frac{1}{N_t}\sum_{i=1}^{N_t}(1-d_i)\log(1-z_i) \tag{12}$$

$$L_{\text{glo}_2} = -\frac{1}{N_s}\sum_{i=1}^{N_s} d_i \log(u_i) - \frac{1}{N_t}\sum_{i=1}^{N_t}(1-d_i)\log(1-u_i). \tag{13}$$

2) Instance-Level Alignment: Similar to feature-level alignment, the UG strategy is also applied to instance-level features. The instance-level feature refers to the regions of interest generated by the detector before input into the final class classifiers. Aligning the instance-level features can help the model better adapt to variations of object instances in different domains, such as the viewpoint and object size. Specifically, the instance-level domain discriminator D_{ins} is designed to focus on the instance-level features. Define the output of D_{ins} for the jth region proposal in the ith images as $(p_i)_j = D_{\text{ins}}(R_{\text{da}}(G_{\text{da}}(x_i)))$. Similarly as above, the uncertainty is computed from D_{ins} as $(o_i)_j = E((p_i)_j)$. Based on computed uncertainty, the domain prediction probability on each region proposal is further computed as $(r_i)_j = (p_i)_j \times (2-(o_i)_j)$. Then, the loss function of the instance-level alignment can be written as

$$L_{\text{ins}} = -\frac{1}{N_s}\sum_{i=1}^{N_s}\sum_{j} d_i \log\left(\left(r_i^s\right)_j\right) - \frac{1}{N_t}\sum_{i=1}^{N_t}\sum_{j}(1-d_i)\log\left(1-\left(r_i^t\right)_j\right). \tag{14}$$

By applying the UG strategy to feature-level and instance-level alignment, the distributional differences between different domains are reduced while preserving the discriminative semantics of the ship targets. This greatly mitigates the negative effects introduced by the adversarial domain adaptation learning strategy in SAR images and further improves the SAR cross-domain ship detection performance.

E. Overall Objective

The overall goal of the proposed SHSC method comprises four parts: the OD objective $L_{\text{det}}^{\text{da}}$ and $L_{\text{det}}^{\text{sa}}$, the SA semantic calibration objective L_{sscm}, and domain adaptation objective L_{adv}. Specifically, L_{sscm} is a composite of different levels of semantic calibration loss, which can be expressed as

$$L_{\text{sscm}} = \underbrace{L_{\text{ssim}_1} + L_{\text{ssim}_2} + L_{\text{ssim}_3}}_{L_{\text{feat}}} + L_{\text{logit}}. \tag{15}$$

Moreover, L_{adv} is a combination of multilevel adversarial training losses, which can be expressed as

$$L_{\text{adv}}(G, D) = L_{\text{loc}} + L_{\text{glo1}} + L_{\text{glo2}} + L_{\text{ins}} \tag{16}$$

where L_{loc}, L_{glo1}, L_{glo2}, and L_{ins} are the losses of the domain discriminator D_l, D_{g1}, D_{g2}, and D_{ins}, respectively, as discussed in Section III-D. Finally, the overall training loss can be formulated as

$$\max_{D}\min_{G,R} L_{\text{det}}^{\text{da}}(G, R) + \alpha L_{\text{det}}^{\text{sa}}(G, R) + \beta L_{\text{sscm}}(G) + \gamma L_{\text{adv}}(G, D) \tag{17}$$

where α, β, and γ are the tradeoff parameters that balance our newly added semantic calibration, domain adaptation components, and the faster R-CNN loss.

F. Theoretical Insights

In this section, we introduce transfer learning theory to further explain the validity of our proposed SHSC. Based on the theory of DA bounds [61], the expected error on the target domain is as follows:

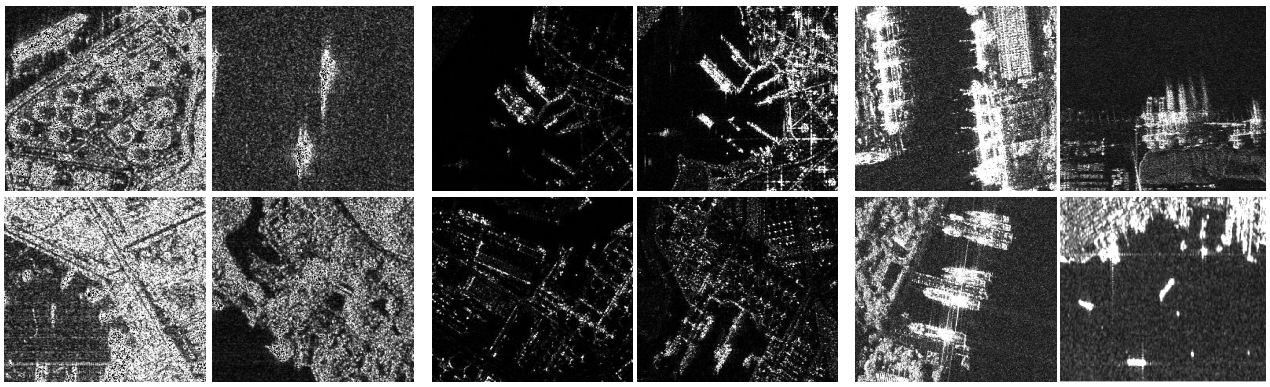
$$\epsilon_T(h) \le \epsilon_S(h) + d_{\mathcal{H}\Delta\mathcal{H}}(\mathcal{D}_S, \mathcal{D}_T) + \lambda \tag{18}$$

where $\epsilon_S(h)$ and $\epsilon_T(h)$ represent errors on the source and target domains, respectively, based on the hypothesis $h \in \mathcal{H}$. $d_{\mathcal{H}\Delta\mathcal{H}}(\mathcal{D}_S, \mathcal{D}_T)$ represents the difference between the source and target domains. λ typically accounts for additional errors due to inherent domain differences. Based on the above generalization boundary theory, we explore the potential of our SHSC to reduce generalization risk. As mentioned above, cross-domain SAR ship detection tasks face unique challenges due to the similarity between ship targets and background, which can lead to the collapse of discriminative representations learned from the source domain. This leads to an increase in $\epsilon_S(h)$, which in turn leads to an increase in $\epsilon_T(h)$. To address these issues, our article introduces two strategies. First, SSCM introduces additional SA detectors and uses KD for semantic calibration of DA representations. This calibration is performed at the feature level (via SSIM loss) and at the logit level (via KL divergence). SSIM loss facilitates the transfer of structural information from the source domain and provides the basis for subsequent logit-based KD. Logit-based KD ensures consistency in the probability distribution of the output layer, further enhancing the discriminative learning of the model. By complementing each other, the discriminative semantics of the target is effectively enhanced. This results in $\epsilon_S(h)$ being reduced, which in turn reduces $\epsilon_T(h)$. Second, UG-FAM and UG-IAM are introduced. For the target detection task, the joint distribution of training samples in the source and target domains can be defined as $P_S(C, B, I)$ and $P_T(C, B, I)$. Due to the domain shift, $P_S(C, B, I) \neq P_T(C, B, I)$. Using Bayes' formula, the joint distribution can be decomposed as $P(C, B, I) = P(C, B|I)P(I)$, where

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TABLE I
COMPARISON OF TECHNICAL PARAMETERS BETWEEN GAOFEN-3, RADARSAT-2, SENTINEL-1, AND TERRASAR-X

Technical Parameters	Gaofen-3	RadarSat-2	Sentinel-1	TerraSAR-X
Orbit altitude	755 km	798 km	693 km	514.8 km
Wavelength	C-band(5.63cm)	C-band(5.63cm)	C-band(5.63cm)	X-band(3.11cm)
Antenna size	15 m × 1.5 m	15 m × 1.37m	12.3 m × 0.821 m	4.8 m × 0.7 m
Peak power	1.5 kW	1.27 kW	4.7 kW	1.8 kW
Incident angle	$10° - 60°$	$20° - 49°$	$20° - 46°$	$20° - 55°$
Signal bandwidth	240 MHz	100 MHz	100 MHz	150 MHz/300MHz
Imaging swath	10-650 km	20-50km	20-400 km	5-100 km
Resolution (Rg. × Az.)	1m × 1m	0.8m × 3m	5m × 5m	3m × 3m
Polarization	Single/Dual/Quad	Single/Dual/Quad	Single/Dual	Single/Dual/Quad
Lifespan	8 year	7.5year	7.5year	5 year



(a) (b) (c)

Fig. 5. Sample images in the three experiment datasets. (a) Gaofen-3. (b) Sentinel-1. (c) SSDD.

$P(C, B|I)$ is the same for both domains as they share the same label space. Thus, domain shifts are primarily due to differences in image representations, $P(I)$. The feature level distributions across domains are aligned, focusing on the local patch and global image features. Local patch features capture local detail differences, while global features emphasize global consistency, thus reducing domain discrepancy. Alternatively, $P(C, B, I)$ can be decomposed as $P(C|B, I)P(B, I)$, where $P(C|B, I)$ is consistent across domains. Therefore, the domain shift results from differences in $P(B, I)$. The instance-level distributions are aligned across domains. This results in the $d_{\mathcal{H}\Delta\mathcal{H}}(\mathcal{D}_S, \mathcal{D}_T)$ can be minimized, and then $\epsilon_T(h)$ is also reduced. With respect to the inequality (18), these changes provide a principled explanation and guarantee the validity of the proposed method. We will demonstrate these changes in the experimental Section IV.

IV. EXPERIMENTS

In this section, we evaluate the proposed SHSC method. The dataset, evaluation criteria, and implementation details are presented in Section IV-A. Comparisons with other state-of-the-art cross-domain detection approaches are reported in Section IV-B. Further empirical analysis and discussion of the model are presented in Section IV-C. More details are provided below.

A. Experiments Setup

1) Datasets: In this article, we build a cross-domain SAR image dataset acquired by RadarSat-2, Gaofen-3, TerraSAR-X, and Sentinel-1 satellites to assess the effectiveness of the proposed SHSC. These SAR images have wide practical applications, such as enabling monitoring of ship activities, shipping lane conditions, and marine traffic density. It is invaluable for marine traffic, resource management, and fishery surveillance. Furthermore, it can detect illegal activities like piracy and smuggling, thus maintaining maritime security and stability. In addition, it can be used for marine environmental monitoring and disaster early warning, providing critical support for maritime rescue and emergency response. However, these SAR platforms exhibit different technical parameters during the imaging process, such as orbital altitude, bandwidth, and incident angle, as shown in Table I, resulting in different distributions of acquired SAR images. Based on these SAR images, we construct three datasets: Gaofen-3, Sentinel-1, and SSDD, abbreviated as G, S, and SSDD for brevity, respectively. The several example samples for each dataset are shown in Fig. 5.

1) **Gaofen-3** dataset is acquired by the Gaofen-3 satellite, which employs C-band multipolarization SAR technology. The wide-format imaging capability of Gaofen-3 enables it to cover wide maritime regions, providing important support for maritime monitoring and management. It can also help to accurately identify the target types, thus improving the efficiency of maritime management. In this article, we use the data collected and generated by Wang et al. [62] for the cross-domain detection task, as shown in Fig. 5(a). The 2495 images with the size 256 × 256 are used for model evaluation,

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of which 1980 images (4963 ships) are utilized for training, and 515 images (1499 ships) are utilized for testing.

2) **Sentinel-1** dataset is derived from the Sentinel-1 satellite, an Earth observation satellite in the European Space Agency's Copernicus program equipped with C-band SAR sensors. The SAR images acquired by Sentinel-1 are widely used in global environmental monitoring, marine research, disaster management, and other fields. Its rapid revisiting capabilities enable it to monitor global environmental changes in real-time. It also has important applications in marine oil pollution and iceberg drift monitoring. Here, we use the data collected and generated by Wang et al. [62], as shown in Fig. 5(b). The 2498 images with the size 256 × 256 are used to assess the model, of which 1980 images (2636 ships) are utilized for training, and 518 images (747 ships) are utilized for testing.
3) **SSDD** dataset contains images from the Sentinel-1, TerraSAR-X, and RadarSat-2 satellites. TerraSAR-X is known for its extremely high resolution and ability to penetrate clouds, giving it a unique advantage in topographic surveying and ground surface deformation monitoring. Its X-band SAR technology can obtain clear and accurate radar images, providing high-quality data for map-making. Meanwhile, through continuous monitoring of surface deformation, TerraSAR-X can detect geological disaster risks in time and provide important information for disaster early warning and prevention. The RadarSat-2 satellite, due to its flexible imaging modes and powerful data processing capability, has been widely used in urban planning, environmental monitoring, disaster assessment, and military reconnaissance. Its multipolarization capability can help extract rich ground surface information, thus supporting environmental monitoring and disaster assessment efforts. In this article, we utilize the data collected by Zhang et al. [63], which contains 1160 images of various image sizes. As shown in Fig. 5(c). The 928 images (2041 ships) are applied for training, and 232 images (546 ships) are applied for testing.

Note that the training and test sets above are divided in an approximately 4:1 ratio, with image selection entirely random to ensure objectivity and unbiased data.

2) Evaluation Criteria: This article utilizes precision, recall, $F1$-score, and average precision (AP) as quantitative evaluation metrics for assessing the performance of the cross-domain detection model. Intersection over union (IoU) measures the area of overlap between the model-predicted bounding box and the ground-truth box. In all experiments, the value of IoU empirically is set to 0.5. This is consistent with most existing SAR target detection approaches. We also present the detection results with various IoU thresholds in Section IV-C. Positive examples represent the ship targets. The negative examples represent the background. The precision measures the proportion of correctly detected ships among all detected ships, defined as

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{19}$$

where TP denotes true positive, indicating the number of positive examples correctly predicted as ship targets. The FP denotes false positives, indicating the number of negative examples incorrectly detected as ship targets. The recall measures the proportion of all ground truth ships that are correctly detected. It is defined as

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{20}$$

where FN denotes false negative, i.e., positive examples misclassified as negative examples. The AP computes the precision and recall values under various thresholds and averages these values to obtain the final AP value, following the evaluation criteria of Pascal VOC2012. Defined as

$$\text{AP} = \int_0^1 P(r)dr \tag{21}$$

where $P(r)$ denotes the precision at recall r. The $F1$-score calculates the harmonic mean of precision and recall for a comprehensive evaluation of model performance

$$F1\text{-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{22}$$

Precision, recall, $F1$-score, and AP values all range from [0, 1]. Higher precision means more accurate predictions and higher recall means the model can localize the target more comprehensively. Higher AP and $F1$ values indicate a greater overall improvement in precision and recall. Using these metrics, the performance of the cross-domain SAR ship detection method can be synthetically evaluated.

3) Implementation Details: The proposed SHSC follows the setting of the cross-domain detection model [52] and uses the faster R-CNN with ResNet-101 [22] pretrained on ImageNet [64]. The source domains in our experiments are fully annotated with bounding boxes and corresponding categories, while the target domains are fully unlabeled. Test performance on the target domain is reported. The shorter side of each image is resized to 256 in all experiments. During each iteration, one image from the source domain and one image from the target domain are sent to the model as a batch, i.e., the batch size is set to 2. The SGD with a momentum of 0.9 is used for optimization, starting with a learning rate of 0.001 and then decreasing to 0.0001 after 30k iterations. The whole model is trained for 50k iterations on an NVIDIA GeForce RTX 4090 GPU. For the domain adaptation tasks between G and S and between S and SSDD, the tradeoff parameter β in (17) is set to 0.1. In all experiments, α and γ are all set to 1, and the τ in (5) is set to 1. In the detection part, the hyperparameter is set as follows [23]. All experiments are implemented in the PyTorch framework [65].

B. Comparisons With State-of-the-Art

To show the superiority of the proposed SHSC method, we compare SHSC with ten state-of-the-art approaches. These include traditional deep learning-based detection algorithms:

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TABLE II
RESULTS ON ADAPTATION FROM GAOFEN3 TO SENTINEL-1 AND SENTINEL-1 TO GAOFEN3

Method	G → S				S → G			
	Precision(%)	Recall(%)	F1-Score(%)	AP(%)	Precision(%)	Recall(%)	F1-Score(%)	AP(%)
Source Only	79.96	84.47	82.15	65.15	85.27	76.78	80.80	65.45
DDQ	81.47	84.61	83.01	65.92	80.36	74.40	77.27	59.70
DINO	81.25	83.73	82.47	67.28	81.68	72.34	76.73	58.55
DAF	82.80	85.68	84.22	70.68	86.64	73.45	79.50	63.22
HTCN	83.67	85.41	84.53	71.65	84.78	79.72	82.17	67.65
SWDA	83.86	86.75	85.28	69.92	88.88	81.32	84.93	70.26
SCL	83.74	86.08	84.98	70.54	86.72	79.19	82.78	69.25
ILLUME	83.69	86.21	84.93	70.16	87.92	77.98	82.65	66.51
TIA	86.01	86.32	86.16	72.48	**89.30**	79.05	83.86	70.08
PT-MAF	85.33	85.81	85.57	72.09	88.50	77.79	82.80	68.36
SHSC(Ours)	**89.76**	**88.22**	**88.98**	**75.71**	87.66	**85.86**	**86.75**	**73.67**

TABLE III
RESULTS ON ADAPTATION FROM GAOFEN3 TO SSDD AND SSDD TO GAOFEN3

Method	G → SSDD				SSDD → G			
	Precision(%)	Recall(%)	F1-Score(%)	AP(%)	Precision(%)	Recall(%)	F1-Score(%)	AP(%)
Source Only	88.75	67.58	76.73	59.92	85.71	69.98	77.05	60.01
DDQ	85.23	65.80	74.27	57.00	85.53	71.74	78.03	61.90
DINO	83.41	62.14	71.22	55.20	79.87	68.39	73.69	55.10
DAF	91.36	70.33	79.48	63.98	85.62	75.72	80.37	64.08
HTCN	88.35	72.71	79.77	63.81	89.30	76.58	82.45	65.94
SWDA	91.95	72.89	81.32	66.26	81.72	72.45	76.81	60.68
SCL	89.10	74.91	81.39	66.51	87.11	74.25	80.17	62.51
ILLUME	90.10	74.36	81.48	66.77	86.91	78.99	82.76	66.08
TIA	90.76	72.53	80.63	65.40	85.65	79.85	82.65	67.25
PT-MAF	88.40	73.01	79.97	65.17	88.17	77.79	82.66	66.97
SHSC(Ours)	**92.86**	**75.82**	**83.48**	**68.20**	**90.59**	**83.19**	**86.73**	**71.77**

TABLE IV
RESULTS ON ADAPTATION FROM SENTINEL-1 TO SSDD AND SSDD TO SENTINEL-1

Method	S → SSDD				SSDD → S			
	Precision(%)	Recall(%)	F1-Score(%)	AP(%)	Precision(%)	Recall(%)	F1-Score(%)	AP(%)
Source Only	84.57	60.44	70.50	52.48	77.48	75.50	76.48	57.77
DDQ	78.69	59.37	67.68	48.91	77.18	77.61	77.39	58.31
DINO	76.56	55.71	64.49	47.84	74.51	71.78	73.12	53.20
DAF	89.80	65.20	75.55	59.23	80.43	76.71	78.53	59.26
HTCN	87.38	64.65	74.32	58.64	79.57	79.92	79.74	60.90
SWDA	87.82	65.38	74.96	59.56	82.74	78.31	80.46	62.29
SCL	87.62	66.12	75.37	58.39	80.13	**81.51**	80.81	61.44
ILLUME	89.22	65.20	75.34	60.01	81.30	78.85	80.0	62.58
TIA	86.44	65.49	74.52	60.42	82.48	77.38	79.85	62.69
PT-MAF	88.99	65.57	75.51	58.16	79.92	77.92	78.91	61.34
SHSC(Ours)	**91.03**	**67.58**	**77.57**	**62.85**	**84.72**	78.58	**81.53**	**66.23**

faster-RCNN [23], DDQ [66], DINO [67], and cross-domain detection algorithms: DAF [49], HTCN [52], SWDA [54], SCL [53], ILLUME [51], TIA [68], and PT-MAF [69]. Notably, the traditional deep learning-based detection algorithms are trained using only source domain data. Once trained, the model is directly used to evaluate the target domain without any domain adaptation. Here, faster-RCNN is used as the baseline and is referred to as the source only in all subsequent experiments. All algorithms employ Resnet101 as the backbone network. The evaluation involves quantitative metrics, including precision, recall, $F1$-score, and AP, and qualitative assessment by visualizing detection results across various methods. Detailed experimental results are provided in the following sections.

1) Quantitative Evaluation: Tables II–IV present the results of cross-domain detection for various algorithms across different tasks, respectively. Table II reports the adaptation results between G and S. For the G→S task, the proposed SHSC demonstrates the best performance across all metrics. For S→G, SHSC also excels in recall, $F1$-score, and AP. Notably, TIA excels in precision because it performs domain alignment by independently optimizing task-specific losses in

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both classification and localization spaces, thereby enhancing detection performance on the target domain. For the S→G task, DAF performs poorly in the recall, $F1$-score, and AP, likely due to its single-layer domain alignment strategy, which fails to effectively bridge the domain gaps, resulting in limited detection capabilities in the target domain. In Table III, SHSC consistently exhibits the best detection performance in the G→SSDD and SSDD→G tasks. The adaptation results between S and SSDD are presented in Table IV. For the S→SSDD task, SHSC outperforms all other methods across all metrics. In the SSDD→G task, SHSC achieves the best performance in precision, $F1$-score, and AP, while SCL excels in the recall, benefiting from its strategy of leveraging target domain information at various stages to adapt the model effectively. Overall, our SHSC method demonstrates robust and effective cross-domain detection capabilities, effectively mitigating the challenges posed by variations in imaging conditions and platforms across different spaceborne SAR images. By preserving the discriminative of ship targets, SHSC effectively bridges the domain gaps, leading to better performance across various tasks. These impressive cross-domain detection results further validate the robustness and effectiveness of our proposed SHSC method.

2) Qualitative Evaluation: Figs. 6–8 present the visualization of detection results obtained by various algorithms on the target domain, respectively. Within these figures, (i) are the ground truths of SAR ship targets, marked by the green rectangles. (ii)–(xii) are the ship targets predicted by the source only, DDQ, DINO, DAF, HTCN, SWDA, SCL, ILLUME, TIA, PT-MAF, and the proposed SHSC, respectively. Here, yellow rectangles indicate targets predicted by the model, blue rectangles indicate targets that are missed detections, and orange ovals indicate targets that are false alarms. In Fig. 6, (a) displays the results from G to S adaptation, and (b) presents the results from S to G adaptation. From Fig. 6, it can be observed that there is a significant distribution difference between G and S. However, the proposed SHSC still exhibits better detection accuracy on the target domain. In the complex inshore scene (rows 1, 3, 5, and 7), SHSC still accurately locates the targets, despite the high visual similarity between the ship targets and the background interference. In contrast, other methods exhibit obvious false and missed detections. Fig. 7 exhibits the visualization results for the adapting between G and SSDD. (a) Displays the results from G to SSDD, and (b) shows the results from SSDD to G. It is apparent from Fig. 7 that SHSC achieves relatively accurate target localization in both scenarios, multitarget and complex inshore scenes. In contrast, other methods suffer from significant missed detections. Fig. 8 illustrates the visualization results for the adapting between S and SSDD. (a) Shows the results for S to SSDD, and (b) shows the results for SSDD to S. As evident from Fig. 8, SHSC is capable of accurately detecting more targets in multitarget scenarios with inshore scenes (rows 1–4). Against a complex background (rows 5–8), SHSC can accurately locate targets without interference from complex backgrounds. This further validates the effectiveness of our proposed method in handling complex and diverse SAR images. Overall, the proposed SHSC demonstrates impressive detection accuracy across various tasks. This further demonstrates the effectiveness and robustness of the proposed SSCM and UG strategies, which effectively preserve and enhance the discriminative features of ship targets during the domain adaptation process. Consequently, the cross-domain detection performance is significantly improved.

C. Further Empirical Analysis

This section provides an in-depth analysis and discussion of the proposed SHSC. All experiments are conducted on G to S tasks. The details are explained below.

1) Ablation Study: The ablation experiments are performed to assess the effectiveness of each component in the SHSC. The results are presented in Table V. In our proposed SHSC, SSCM is introduced, which uses the source representation as a guide for semantic calibration to domain adaptation features through feature-based KD and logit-based KD. To evaluate their effectiveness, we remove them from the model and notate them as Ours w/o L_{logit}, Ours w/o L_{feat}, and Ours w/o SSCM (i.e., without L_{logit} and L_{feat}), respectively. From Table V, it can be observed that Ours w/o SSCM brings considerable degradation in model performance, with a drop by -3.49% (precision), -1.81% (recall), -2.64% ($F1$-score), and -3.70% (AP). Although the introduction of SSIM can significantly improve the performance of the model, it inevitably leads to an increase in the number of model parameters. Furthermore, Ours w/o L_{logit} and Ours w/o L_{feat} also lead to performance degradation. These results indicate that, for SAR images, applying multilevel semantic calibration in the domain adaptation process can effectively preserve the discriminative information, thus facilitating cross-domain detection accuracy. In addition, the UG strategy is introduced, which operates on the local patch feature alignment (i.e., L_{loc}) and global feature alignment (i.e., L_{glo}), and instance-level alignment (i.e., L_{ins}), respectively. Similarly, to assess their contributions, we remove them from the model and denote them as Ours w/o L_{loc}, Ours w/o L_{glo}, and Ours w/o L_{ins}, respectively. From Table V, it can be seen that removing these components can lead to an obvious degradation in the cross-domain detection performance. Finally, the full model shows the best performance. This suggests the effectiveness of the proposed components in improving both the adaptability and the discriminability of the model, each of which is indispensable.

2) Effectiveness of UG: In the proposed method, based on DAT, the UG strategy is introduced, which operates at both the feature and instance levels for aligning the domain distributions. The primary purpose of UG is to force the model to adequately extract the discriminative representations of ship targets. To validate the effectiveness of the UG strategy, we first remove the SSCM from the model, and then the impact of UG is evaluated by comparing the model without the UG, i.e., Ours w/o UG, and the model with the UG, i.e., Ours with UG. The experimental results are presented in Fig. 9. From Fig. 9, it can be observed that introducing the UG strategy obviously improves the cross-domain detection performance. Moreover, we visualize the class activation maps of the output feature maps from the last convolutional layer of the backbone network, as presented in Fig. 10. These class activation maps

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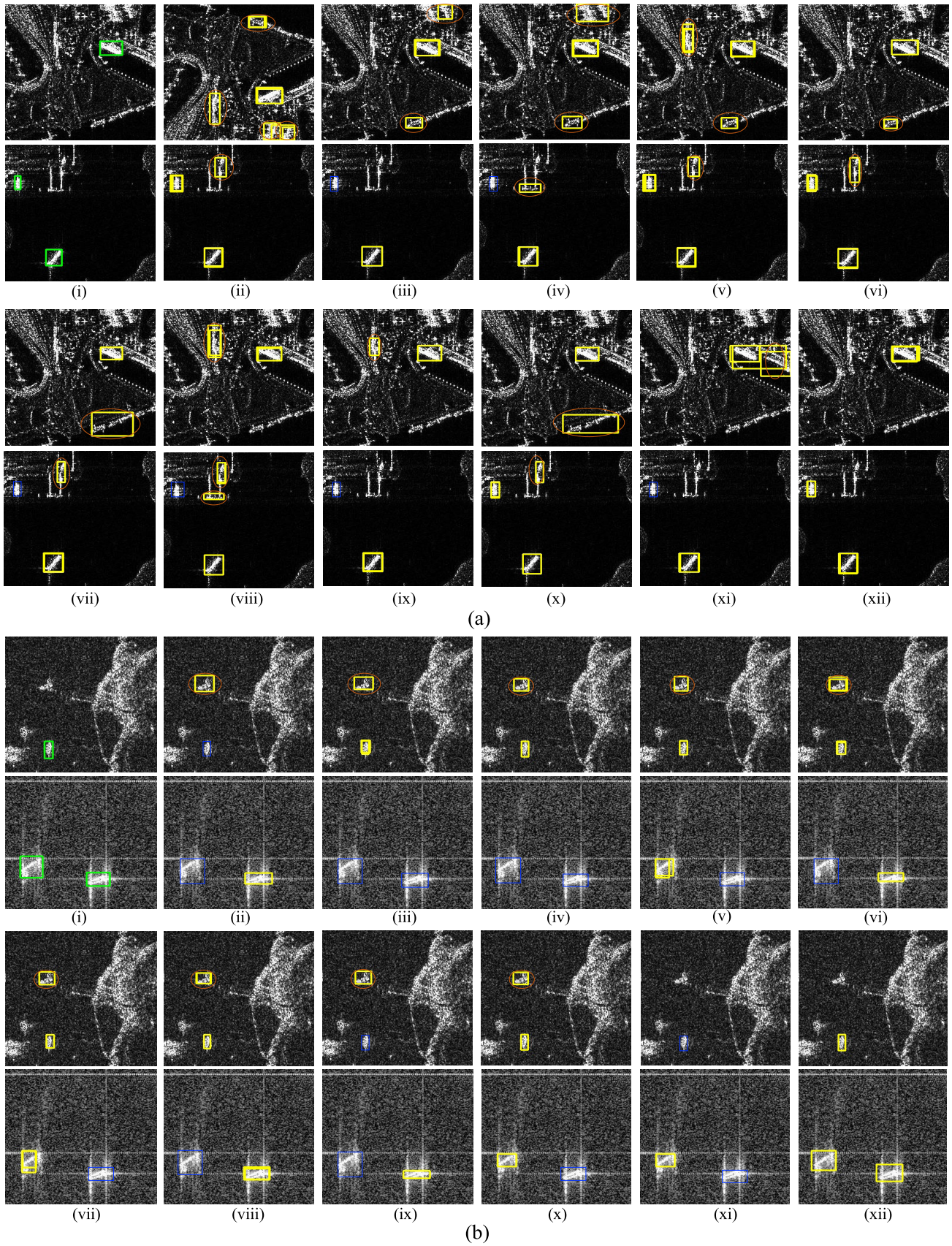


Fig. 6. Illustration of the detection results obtained by various algorithms on the target domain. (a) Results from G to S adaptation, and (b) results from S to G adaptation. (i) Ground truth. (ii) Source only. (iii) DDQ. (iv) DINO. (v) DAF. (vi) HTCN. (vii) SWDA. (viii) SCL. (ix) ILLUME. (x) TIA. (xi) PT-MAF. (xii) SHSC (Ours). Green rectangles mark the ground truths of SAR ship targets. Yellow rectangles indicate targets predicted by the model, blue rectangles indicate targets that are missed detections, and orange ovals indicate targets that are false alarms.

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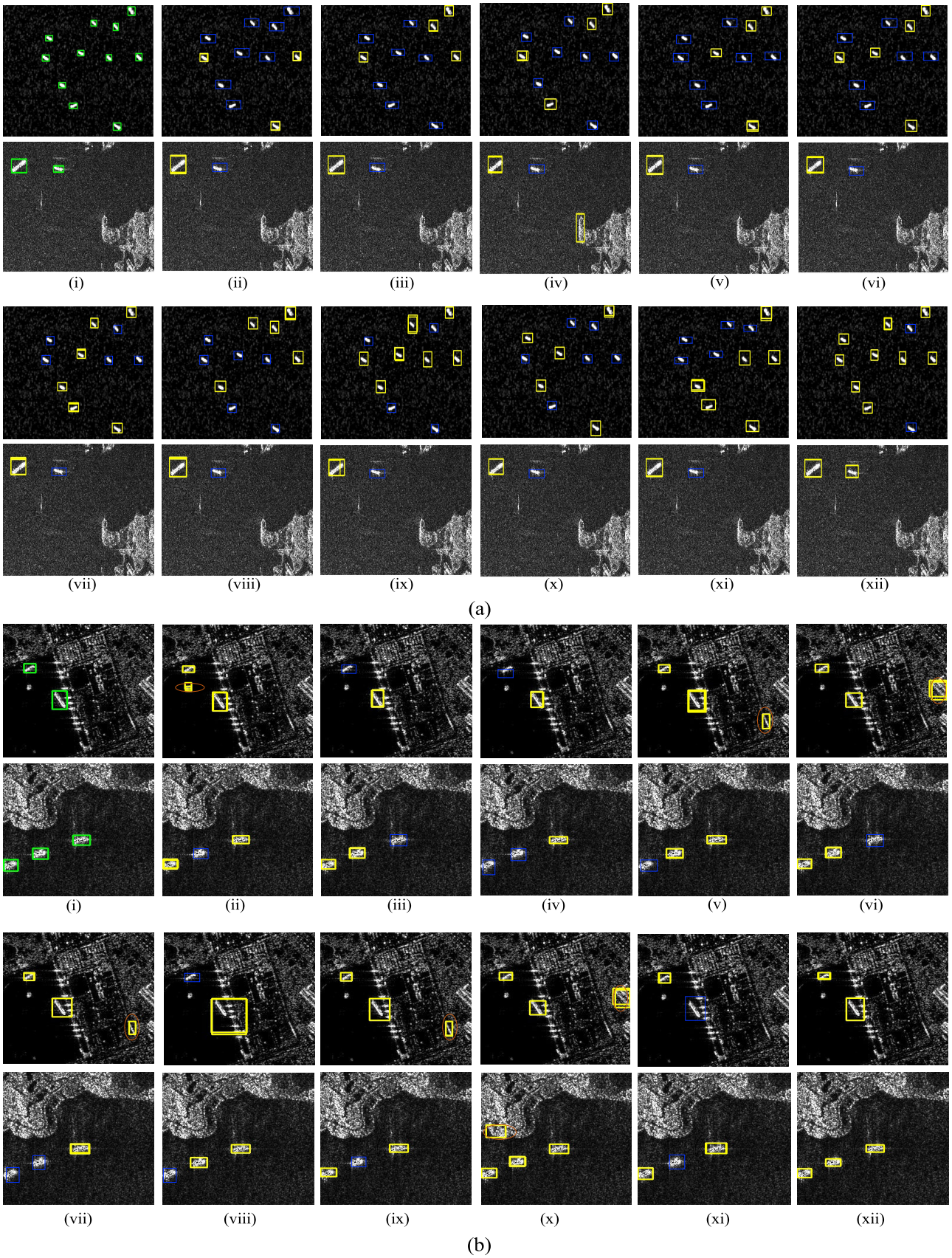


Fig. 7. Illustration of the detection results obtained by various algorithms on the target domain. (a) Results from G to SSDD adaptation, and (b) results from SSDD to G adaptation. (i) Ground truth. (ii) Source only. (iii) DDQ. (iv) DINO. (v) DAF. (vi) HTCN. (vii) SWDA. (viii) SCL. (ix) ILLUME. (x) TIA. (xi) PT-MAF. (xii) SHSC (Ours). Green rectangles mark the ground truths of SAR ship targets. Yellow rectangles indicate targets predicted by the model, blue rectangles indicate targets that are missed detections, and orange ovals indicate targets that are false alarms.

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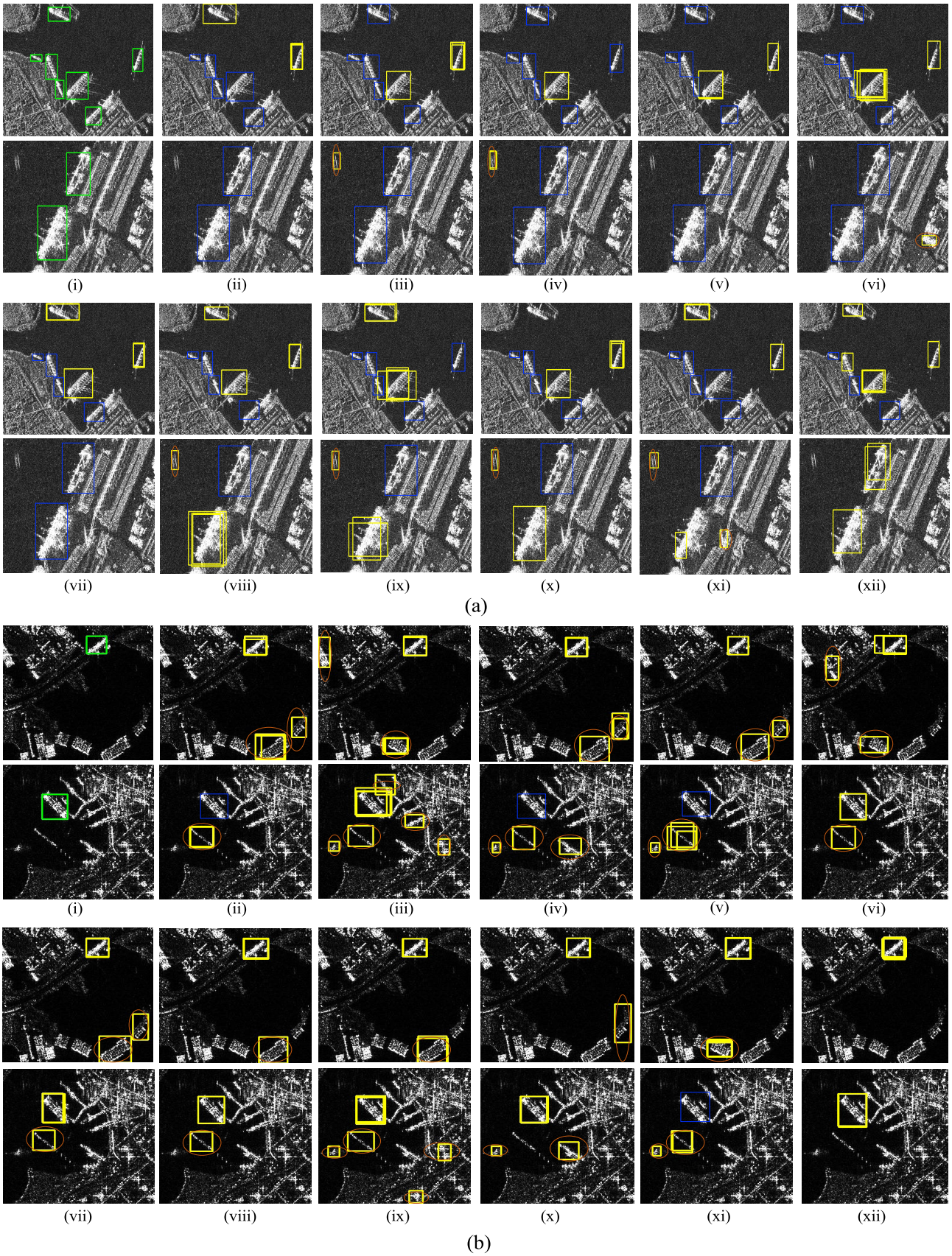


Fig. 8. Illustration of the detection results obtained by various algorithms on the target domain. (a) Results from S to SSDD adaptation, and (b) results from SSDD to S adaptation. (i) Ground truth. (ii) Source only. (iii) DDQ. (iv) DINO. (v) DAF. (vi) HTCN. (vii) SWDA. (viii) SCL. (ix) ILLUME. (x) TIA. (xi) PT-MAF. (xii) SHSC (Ours). Green rectangles mark the ground truths of SAR ship targets. Yellow rectangles indicate targets predicted by the model, blue rectangles indicate targets that are missed detections, and orange ovals indicate targets that are false alarms.

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TABLE V
ABLATION STUDIES OF DIFFERENT COMPONENTS ON G TO S TASK

Method	L_{det}^{sa}	L_{feat}	L_{logits}	L_{loc}	L_{glo}	L_{ins}	Precision(%)	Recall(%)	F1-Score(%)	AP(%)	Param.(M)(%)
Source Only	✘	✘	✘	✘	✘	✘	79.96	84.47	82.15	65.15	46.93
Ours w/o SSCM	✘	✘	✘	✔	✔	✔	86.27	86.41	86.34	72.01	56.02
Ours w/o L_{logits}	✔	✔	✘	✔	✔	✔	88.89	86.61	87.74	74.75	103.18
Ours w/o L_{feat}	✔	✘	✔	✔	✔	✔	87.79	85.94	86.86	72.59	103.18
Ours w/o L_{loc}	✔	✔	✔	✘	✔	✔	85.03	87.01	86.01	72.22	103.07
Ours w/o L_{glo}	✔	✔	✔	✔	✘	✔	87.59	86.75	87.17	72.94	94.62
Ours w/o L_{ins}	✔	✔	✔	✔	✔	✘	88.03	86.61	87.31	73.89	102.96
Ours	✔	✔	✔	✔	✔	✔	**89.76**	**88.22**	**88.98**	**75.71**	103.18

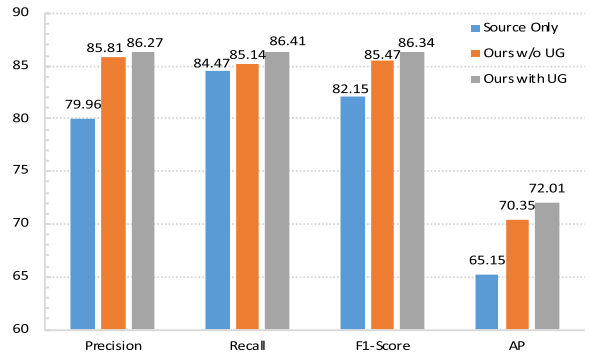


Fig. 9. Evaluation of the UG strategy.

TABLE VI
EFFECT OF THE DIFFERENCE MEASURE FUNCTIONS

Method	Precision(%)	Recall(%)	F1-Score(%)	AP(%)
Baseline	86.27	86.41	86.34	72.01
ℓ_1	89.12	87.15	88.12	74.72
ℓ_2	86.77	87.01	86.89	73.88
L_{ssim}	**89.76**	**88.22**	**88.98**	**75.71**

reflect the regions of interest for the models. The color's brightness indicates the corresponding region's importance, and the brighter the region, the more it contributes to the model classification. In Fig. 10(a) and (d) represent the ground truth, and (b) and (e) are the results of Ours w/o UG. (c) and (f) are the results of Ours with UG. It can be seen that Ours with UG focuses on image regions that are more relevant to ship targets, which can help the model classify and localize ship targets. This further demonstrates that introducing UG can help the model better extract discriminative representations related to ship targets.

3) Sensitivity of Hyperparameter: In (17), the hyperparameters α, β, and γ are utilized to balance the faster R-CNN loss and the proposed new components $L_{\text{det}}^{\text{sa}}$, L_{sscm}, and L_{adv}, respectively. In (5), τ is the temperature coefficient that controls the smoothness of the softmax output at the logit level during semantic calibration. In this section, the effects of different values of these hyperparameters on the model performance are systematically evaluated, and the results are presented in Fig. 11. Here, the results are recorded by changing one and fixing the others, where the different parameters are initialized to 1. For α, it determines the proportion of detection loss of the SA detector. The larger α value prompts the SA detector to learn the knowledge from the assisted source domain more fully, thus ensuring that the semantic calibration can be effectively implemented. From Fig. 11, it can be seen that different α values inevitably introduce performance fluctuations. Overall, the model's performance is improved with the increase of the α value. Therefore, in all our experiments, to ensure the stability and effectiveness of the model, the α is set to 1. For β, it controls the weight of the semantic calibration loss. The essence of the semantic calibration is to facilitate discriminative learning of domain adaptation representations. Notably, when there is a significant domain shift between the data, reducing β appropriately can mitigate the damage to the model's generalization ability. As illustrated in Fig. 11, the model is sensitive to changes in the β value. This sensitivity arises from the delicate balance struck between semantic calibration and DAT, with β serving as the critical knob that adjusts the tradeoff between model generalizability and discriminability. Combined with Fig. 11, it can be seen that in cross-domain adaptation tasks involving Gaofen-3 and Sentinel-1, the β performs best at 0.1 due to their significant domain differences. For γ, it regulates the proportion of domain adaptation loss, which ensures efficient alignment between the source and target domains. As Fig. 11 illustrates, the model's performance gradually improves as the γ value increases. This shows the effectiveness of DAT for cross-domain adaptation. Therefore, in all of our experiments, the value of γ is set to 1 to ensure efficient domain alignment. For τ, a larger τ leads to a smoother feature distribution, but may potentially impair the discriminative of the features. Since semantic calibration aims to enhance the discriminative of the domain adaptation representations, we tend to choose a relatively small τ. As seen in Fig. 11, the smaller τ can lead to better performance. Therefore, τ is set to 1 in all our experiments. In general, different parameters control the influence of each component on the model. In practice, however, the choice of these parameters also depends on the data distribution. Therefore, selecting appropriate parameters for different data can help the model learn more discriminative and adaptability representations, so as to achieve more stable cross-domain detection.

4) Effect of Difference Measure Function: The proposed method introduces SSCM to perform multilevel semantic calibration on DA detectors through the KD mechanism.

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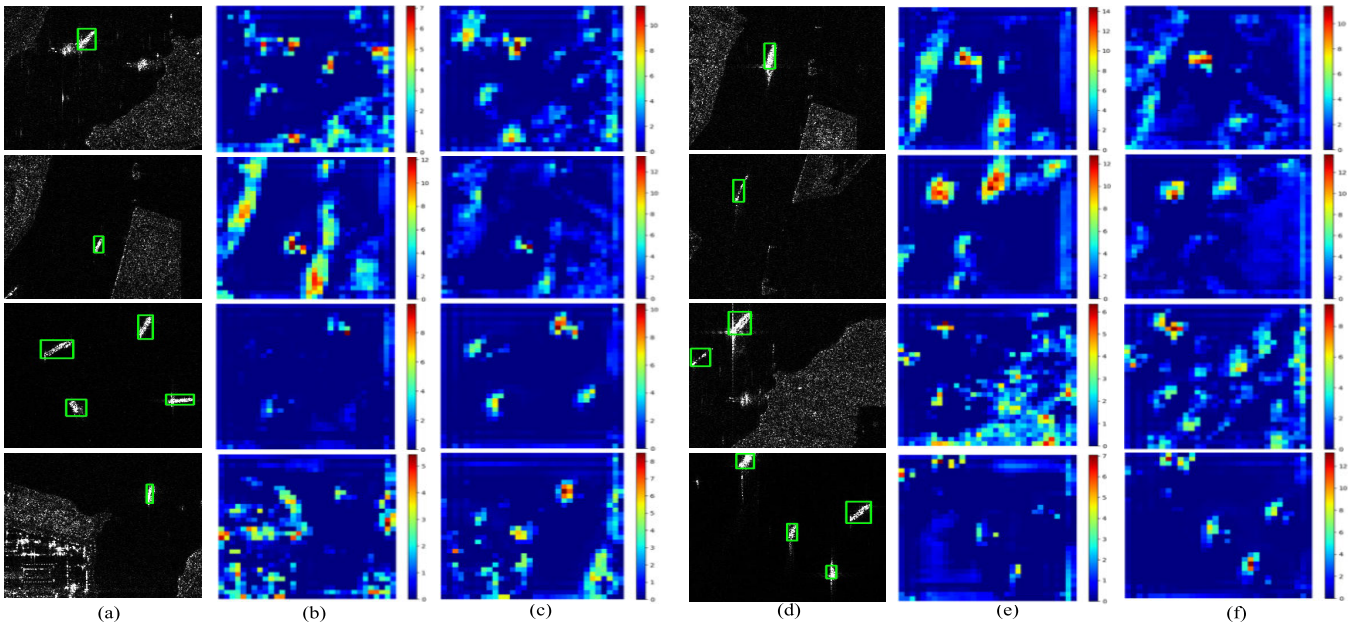


Fig. 10. Visualization of the class activation map derived from the last convolutional layer of the backbone network. (a) and (d) Ground truth. (b) and (e) Results of Ours w/o UG. (c) and (f) Results of Ours with UG.

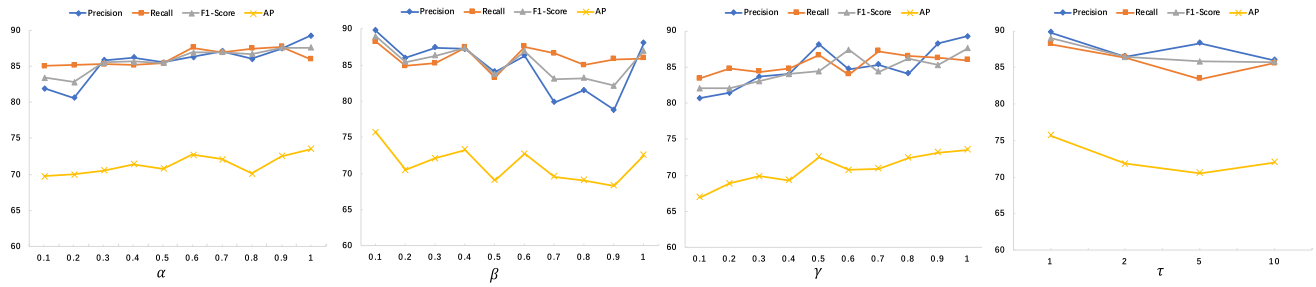


Fig. 11. Influence of different α, β, γ, and τ values on the model performance.

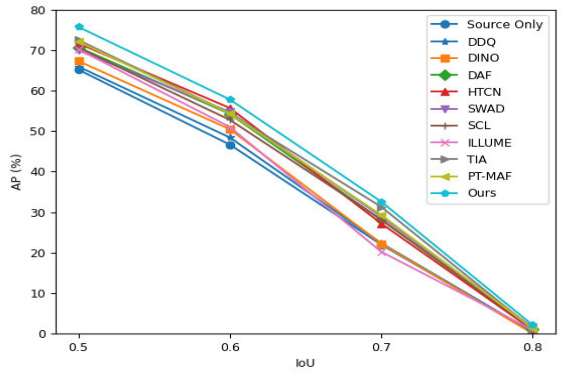
Considering the SAR image characteristics, where actual ship targets usually have high visual similarity with the background environment in texture, their discriminative semantics are mainly reflected in the structural information. Therefore, in order to better preserve the discriminative semantics of ship targets, L_{ssim} is introduced as the difference measure function in SSCM to perform semantic calibration. This section investigates the influence of the difference measure functions, including the ℓ_1 and ℓ_2 functions. The experimental results are presented in Table VI. Here, we remove the SSCM from the model as a baseline. Table VI shows that the various difference measure functions contribute to performance improvements, with SSIM showing a particularly significant improvement, reaching $+3.49\%$ (precision), $+1.81\%$ (recall), $+2.64\%$ ($F1$-score), and $+3.70\%$ (AP). These results indicate that the introduction of L_{ssim} as the difference measure function can help the model to more effectively mine the discriminative semantics related to ship targets, and is further used for semantic calibration during the domain adaptation process, thus improving cross-domain detection performance.

5) Effect of Semantic Calibration on Different Layers: The SA detector is constructed in the proposed method to perform multilevel semantic calibration on domain adaptation features at both the feature and logit levels. In this section, we explore the effect of semantic calibration at various layers on the model performance. The experimental results are reported in Table VII. Here, the baseline represents the model without semantic calibration at the feature level. C1 denotes the model with semantic calibration at the first block of the backbone network, C2 corresponds to the model with semantic calibration at the second block of the backbone network, and so forth for the remaining models. Table VII shows that C1&C2&C3, i.e., the model with simultaneous semantic calibration at all

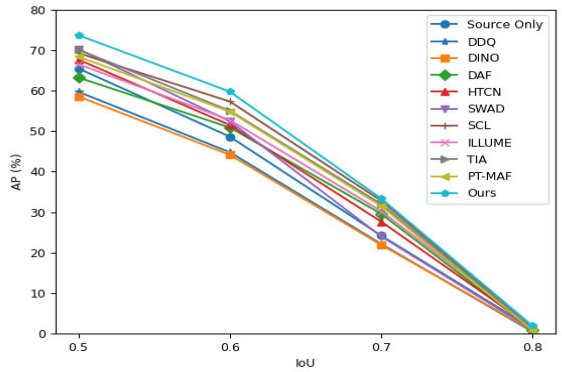
TABLE VII
IMPACT OF SEMANTIC CALIBRATION ON DIFFERENT LAYERS

Method	Precision(%)	Recall(%)	F1-Score(%)	AP(%)
Baseline	87.79	85.94	86.86	72.59
C1	87.98	86.04	87.00	73.28
C2	87.81	86.68	87.24	73.98
C3	87.89	86.81	87.35	73.68
C1&C2	88.47	86.88	87.67	74.82
C1&C3	87.81	87.76	87.78	74.25
C2&C3	88.26	86.48	87.36	74.54
C1&C2&C3	**89.76**	**88.22**	**88.98**	**75.71**

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(a)



(b)

Fig. 12. Detection accuracy with the various IoU thresholds on different adaptation tasks. (a) Results from G to S adaptation and (b) results from S to G adaptation.

levels, performs best in precision, recall, $F1$-score, and AP. This further validates the effectiveness of multilevel semantic calibration. While C1, C2, and C3 models show slight improvements over the baseline, their performance is not as good as the simultaneous multilevel semantic calibration. This implies that the proposed multilevel semantic calibration strategy can help the model more adequately capture the discriminative representation of the target, thus facilitating cross-domain detection performance.

6) Analysis of IoU Threshold: IoU represents the area of overlap between the model-predicted bounding box and the ground-truth box. A higher IoU value indicates a greater overlap area. Setting the IoU to different thresholds can be utilized to assess the detection performance of the model for varying degrees of overlap. This section compares the proposed SHSC with ten other methods across various IoU thresholds on domain adaptation tasks. The experimental results are reported in Fig. 12. (a) Represents the results from G to S adaptation, and (b) represents the results from S to G adaptation. As observed from Fig. 12, the AP value decreases sharply with the increase of the IoU threshold. This is because higher IoU thresholds impose more stringent criteria on the model to locate the object. The proposed SHSC outperforms all other approaches across different thresholds on both domain adaptation tasks. This indicates that the SHSC method exhibits both high detection accuracy and robustness.

7) T-SNE Visualization Feature Map: To explore the influence of various methods on domain features, the T-SNE [70] technology is used to visualize the output features from the last convolutional layer of the backbone of different methods in the S to G adaptation task. This visualization technique offers an intuitive means of assessing feature distributions across domains by mapping them into a 2-D space. If the feature distributions of the two domains overlap significantly in this space, it indicates that they are closely aligned in the feature space, suggesting minimal domain differences. The T-SNE visualization results of different algorithms are shown in Fig. 13. The orange region represents the feature distribution from G. The blue area represents the feature distribution from S. (a)–(k) Represent source only, DDQ, DINO, DAF, HTCN, SWDA, SCL, ILLUME, TIA, PT-MAF, and the proposed SHSC, respectively. From Fig. 13, it can be seen that the traditional deep learning-based detection algorithms, namely, source only, DDQ, and DINO, exhibit obvious differences in feature distribution due to the absence of domain adaptation. In contrast, the UDA-based domain adaptation method can significantly minimize these differences, leading to a more overlapping feature distribution between the source and target domains. In particular, the proposed SHSC, i.e., (k), shows smaller interdomain differences than other methods. This further shows the effectiveness of SHSC in mitigating domain disparities, thereby enhancing cross-domain detection performance in SAR tasks.

8) Efficiency and Computational Costs: This section compares the proposed SHSC with ten other methods regarding model efficiency and computational cost. Specifically, the number of parameters and the inference time are calculated for different models. Inference time is an important metric for measuring the real-time performance of a model. It measures the time required for the model to perform forward propagation on the given input data to produce output predictions, expressed in milliseconds. The results are reported in Table VIII. Table VIII shows that although SHSC achieves the best cross-domain detection performance, the inference time is relatively long. This is mainly due to the introduction of additional SA detectors for semantic calibration. This helps improve detection accuracy but also increases the computational cost accordingly. In contrast, source only, DAF, SWDA, and HTCN performed better regarding the number of model parameters and inference time, showing higher efficiency. However, the cross-domain detection accuracy of these methods is relatively low, and they may be more suitable for specific application scenarios that require high real-time performance but less stringent accuracy. Notably, although the ILLUME effectively improves the structural attention of the object-related region by introducing the self-attention feature map module. However, it also makes the inference time-consuming. In summary, while SHSC performs well in accuracy, it may need to be weighed against its long inference time in application scenarios that pursue high efficiency and low cost. Other methods, such as source only, DAF, SWDA, and HTCN, may be advantageous in applications where real-time performance is critical. Therefore, when selecting the appropriate algorithm, it is necessary to consider the balance between accuracy and computational cost according to the specific application requirements. In the future, we will

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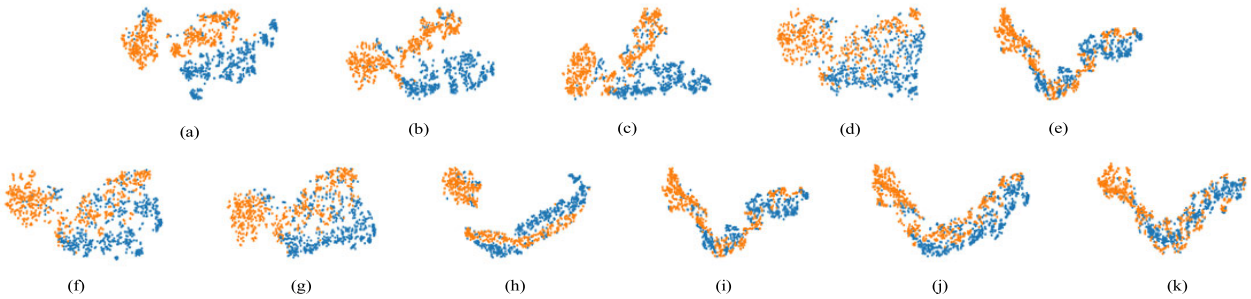


Fig. 13. T-SNE visualization results of different algorithms in the S to G adaptation task. The orange region represents the feature distribution from G. The blue area represents the feature distribution from S. (a) Source only. (b) DDQ. (c) DINO. (d) DAF. (e) HTCN. (f) SWDA. (g) SCL. (h) ILLUME. (i) TIA. (j) PT-MAF. (k) SHSC (Ours).

TABLE VIII
COMPARISON OF DIFFERENT ALGORITHMS IN MODEL EFFICIENCY AND COMPUTATIONAL COST

Method	Source Only	DDQ	DINO	DAF	HTCN	SWDA	SCL	ILLUME	TIA	PT-MAF	SHSC (Ours)
Param.(M)	46.93	67.27	66.37	47.94	55.92	52.71	63.76	66.45	61.31	101.37	103.18
Times (ms/img)	12.20	22.76	25.84	12.40	19.40	17.50	17.30	29.20	21.20	26.40	26.50
AP (%)	65.15	65.92	67.28	70.68	71.65	69.92	70.54	70.16	72.48	72.09	75.71

continue to work on improving the model to achieve a more efficient and higher-performance model.

V. CONCLUSION

This article proposed a SHSC method for ship detection across different spaceborne SAR images. SHSC aims to alleviate the problem of discriminative corruption of ship targets during domain adversarial learning induced by the specificity of SAR images. To this end, the SSCM, the UG-FAM, and the UG-IAM are developed to preserve the discriminative semantics of ship targets during domain adaptation. The SSCM is designed to perform multilevel semantic calibration by introducing additional source representations as guidance. UG-FAM and UG-IAM are developed to enhance the learning of discriminative samples. Compared with other methods, SHSC considers the characteristics of SAR images. It emphasizes enhancing the discriminative learning of ship targets during domain adaptation, resulting in more effective and robust cross-domain detection capabilities. Extensive experiments are conducted on the SAR ship detection datasets acquired by the Gaofen-3, RadarSat-2, Sentinel-1, and TerraSAR satellites. The experiment results consistently demonstrate the superiority of the SHSC approach when dealing with images from different spaceborne SAR platforms. Notably, although SHSC presents better cross-domain detection performance than other methods, it also increases the computational cost accordingly. In the future, we will continue to improve the model to achieve a more efficient and higher detection performance.

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